

User Authentication & Registration (Phase 1)

1. Overview

This document defines the requirements for user authentication, registration, and basic profile management for Phase 1 of the application.

The goal is to enable secure access to the application while ensuring a smooth onboarding experience for first-time users.

2. Scope

Phase 1 includes:

- Application layout (basic navigation structure)
 - Login and logout functionality
 - First-time user registration
 - Basic user profile creation
-

3. Authentication Methods

The system shall support the following login methods:

1. **Gmail SSO**
 2. **Mobile Number Login**
-

4. First-Time User Registration

4.1 Trigger Condition

Registration is required when:

- A user logs in for the first time using Gmail SSO or mobile number
 - The system does not find an existing user record
-

5. Mandatory Registration Fields

The following fields are required during initial registration:

Field	Description	Validation
Mobile Number	User's phone number	Required, unique, valid format
OTP Verification	Required only for Gmail SSO users	Must be successfully verified
Name	Full name of user	Required
Role	User role in system	Required, predefined values
Location	User location	Required

6. Optional Profile Fields (Post-Registration)

These fields are optional and can be completed after registration:

Field	Description	Validation
Farm Size / Type	Small-scale, Mid-scale, etc.	Optional
Crops / Livestock	Types managed by user	Optional, multi-value
Number of Workers	Workforce size	Optional, positive integer
Profile Photo	User image	Optional, JPG/PNG

7. User Flow

7.1 Login Flow

1. User opens application
2. User selects login method:
3. Gmail SSO
4. Mobile number
5. System authenticates user
6. System checks if user exists:
7. If yes → Grant access
8. If no → Redirect to registration

7.2 Registration Flow

1. User enters required details
2. If Gmail login:
3. System sends OTP to mobile number
4. User verifies OTP
5. System validates all fields
6. User account is created

7. User is redirected to application dashboard

7.3 Post-Registration Flow

1. User accesses application
 2. System prompts for profile completion (optional)
 3. User may update additional profile fields anytime
-

8. Validation Rules

- All mandatory fields must be completed
 - Mobile number must be unique
 - OTP verification is mandatory for Gmail SSO users
 - Optional fields can be skipped
-

9. Non-Functional Requirements

- Secure authentication (OAuth for Gmail SSO)
 - OTP delivery reliability
 - Fast response time for login/registration
 - Data privacy and secure storage of user information
-

10. Dependencies

- Gmail OAuth integration
 - OTP service provider
 - Backend user management system
-

11. Success Criteria

- Users can successfully register and log in
- Minimal drop-off during onboarding
- Accurate user data capture
- Seamless transition from registration to application usage